

Architectural Document

Team Yellow “Clarity”

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The purpose of this Business Intelligence solution is to turn raw procurement data into something that is easier to use and more useful for decision-making. This architecture shows how data moves through the system from the point it is collected to how it is finally used in dashboards. The goal is to improve data quality, reduce manual work, and give users access to consistent and reliable information. The system follows a structured flow that includes data sources, data preparation, Tableau processing, and dashboard outputs.

The system starts with procurement data collected from the Virginia Department of Corrections and the Division of Purchasing and Supply. This data includes vendor names, spending amounts, contract details, and SWAM classifications. Although the data is important, it cannot be used right away because it may contain missing values, inconsistent formats, or errors. Earlier in the project, these issues made analysis more difficult and required extra manual work. Because of this, the next step focuses on preparing the data so it can be used correctly.

During the data preparation stage, the dataset is cleaned and organized to make it reliable for analysis. This includes fixing missing or incorrect values, standardizing formats, and creating calculated fields that are needed for reporting. Some of these fields include fiscal year, fiscal quarter, total spend, vendor utilization rate, and SWAM participation rate. The data is also aligned with the VADOC fiscal calendar so that reporting is consistent across all time periods. This step helps ensure that all results are based on the same definitions and reduces the need for repeated corrections.

Once the data is prepared, it is brought into Tableau, which is used to analyze and visualize the data. Tableau allows the team to calculate key metrics and display them in a way that is easy to understand. Users can view total spending trends, SWAM participation, and

vendor utilization across fiscal years and quarters. The system also includes filters so users can focus on specific vendors or time periods. In addition, forecasting is used to give an estimate of future spending based on past trends. Tableau works well for this project because it is easy to use and can handle large datasets while still creating clear visuals.

The dashboards are the final part of the system, where users interact with the data. Instead of working with raw spreadsheets, users can view visual summaries that highlight key information. The dashboards show spending trends, SWAM participation, and vendor activity, and they allow users to filter the data to explore specific details. This makes it easier to understand patterns and supports more informed decision making.

Overall, the system connects each step of the process into one clear flow. Data is collected, prepared, analyzed, and then presented in dashboards that users can interact with. This approach improves reporting speed, increases consistency, and makes the data easier to understand. It also addresses the main challenges identified at the beginning of the project, such as slow reporting, inconsistent metrics, and difficulty accessing clear insights.

In conclusion, this architecture provides a practical solution for managing procurement data. It improves the quality of the data, reduces manual effort, and supports better decision making. The system is designed to be simple to use while still providing meaningful insights, making it useful for both technical and non technical users. By organizing the data into a clear process, the solution helps turn raw information into something that can be used effectively.